Class 08:

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Background

The goal of this mini-project is for you to explore a complete analysis using the unsupervised learning techniques covered in class.

Today we will analyze a biopsy data-set from fine needle aspiration (FNA) of a breast mass.

Data Import

The data is made available as a CSV file for download. We cab read this using `read.csv`

```
fna.data <- "WisconsinCancer (1).csv"
```

```
wisc.df <- read.csv(fna.data, row.names = 1)
```

```
head(wisc.df,3)
```

```
         diagnosis radius_mean texture_mean perimeter_mean area_mean
842302           M       17.99        10.38          122.8      1001
842517           M       20.57        17.77          132.9      1326
84300903         M       19.69        21.25          130.0      1203
         smoothness_mean compactness_mean concavity_mean concave.points_mean
842302           0.11840          0.27760         0.3001             0.14710
842517           0.08474          0.07864         0.0869             0.07017
84300903         0.10960          0.15990         0.1974             0.12790
         symmetry_mean fractal_dimension_mean radius_se texture_se perimeter_se
842302          0.2419                0.07871    1.0950     0.9053        8.589
842517          0.1812                0.05667    0.5435     0.7339        3.398
84300903        0.2069                0.05999    0.7456     0.7869        4.585
         area_se smoothness_se compactness_se concavity_se concave.points_se
842302    153.40      0.006399        0.04904      0.05373           0.01587
842517     74.08      0.005225        0.01308      0.01860           0.01340
84300903   94.03      0.006150        0.04006      0.03832           0.02058
         symmetry_se fractal_dimension_se radius_worst texture_worst
842302       0.03003             0.006193        25.38         17.33
842517       0.01389             0.003532        24.99         23.41
84300903     0.02250             0.004571        23.57         25.53
         perimeter_worst area_worst smoothness_worst compactness_worst
842302             184.6       2019           0.1622            0.6656
842517             158.8       1956           0.1238            0.1866
84300903           152.5       1709           0.1444            0.4245
         concavity_worst concave.points_worst symmetry_worst
842302            0.7119               0.2654         0.4601
842517            0.2416               0.1860         0.2750
84300903          0.4504               0.2430         0.3613
         fractal_dimension_worst
842302                   0.11890
842517                   0.08902
84300903                 0.08758
```

Make sure we remove or exclude the `diagnpsis` column from the data set that we use for further analysis.

```
wisc.data <- wisc.df [,-1]
diagnosis <- as.factor (wisc.df$diagnosis)
```

Exploratory Data Analysis

Q1. How many observations are in this dataset?

```r
nrow(wisc.data)
```

```
[1] 569
```

> Q2. How many of the observations have a malignant diagnosis?

```r
table(wisc.df$diagnosis)
```

```

  B   M 
357 212 
```

So in the dataset, 212 are malignant.

> Q3. How many variables/features in the data are suffixed with `_mean`?

We can use the `grep()` function to help us here:

```r
colnames(wisc.data)
```

```
 [1] "radius_mean"             "texture_mean"           
 [3] "perimeter_mean"          "area_mean"              
 [5] "smoothness_mean"         "compactness_mean"       
 [7] "concavity_mean"          "concave.points_mean"    
 [9] "symmetry_mean"           "fractal_dimension_mean" 
[11] "radius_se"               "texture_se"             
[13] "perimeter_se"            "area_se"                
[15] "smoothness_se"           "compactness_se"         
[17] "concavity_se"            "concave.points_se"      
[19] "symmetry_se"             "fractal_dimension_se"   
[21] "radius_worst"            "texture_worst"          
[23] "perimeter_worst"         "area_worst"             
[25] "smoothness_worst"        "compactness_worst"      
[27] "concavity_worst"         "concave.points_worst"   
[29] "symmetry_worst"          "fractal_dimension_worst"
```

```r
length(grep("_mean", colnames(wisc.data),value = T))
```

```
[1] 10
```

So 10 _mean's are in the data set.

Principal Component Analysis (PCA)

We need to scale our data before PCA with the `scale = TRUE` argument to `prcomp()`.

> Q4. From your results, what proportion of the original variance is captured by the first principal component (PC1)?

PC1 component captures 44.27% of the orignal variance.

> Q5. How many principal components (PCs) are required to describe at least 70% of the original variance in the data?

```
To explain at least 70% of the variance, 3 principal components are required.
```

> Q6. How many principal components (PCs) are required to describe at least 90% of the original variance in the data?

To explain at least 90% of the variance, 7 principles are required.

```r
wisc.pr <- prcomp(wisc.data, scale = TRUE)
summary(wisc.pr)
```

```
Importance of components:
                          PC1    PC2     PC3     PC4     PC5     PC6     PC7
Standard deviation     3.6444 2.3857 1.67867 1.40735 1.28403 1.09880 0.82172
Proportion of Variance 0.4427 0.1897 0.09393 0.06602 0.05496 0.04025 0.02251
Cumulative Proportion  0.4427 0.6324 0.72636 0.79239 0.84734 0.88759 0.91010
                           PC8    PC9    PC10   PC11    PC12    PC13    PC14
Standard deviation     0.69037 0.6457 0.59219 0.5421 0.51104 0.49128 0.39624
Proportion of Variance 0.01589 0.0139 0.01169 0.0098 0.00871 0.00805 0.00523
Cumulative Proportion  0.92598 0.9399 0.95157 0.9614 0.97007 0.97812 0.98335
                          PC15    PC16    PC17    PC18    PC19    PC20   PC21
Standard deviation     0.30681 0.28260 0.24372 0.22939 0.22244 0.17652 0.1731
Proportion of Variance 0.00314 0.00266 0.00198 0.00175 0.00165 0.00104 0.0010
Cumulative Proportion  0.98649 0.98915 0.99113 0.99288 0.99453 0.99557 0.9966
                          PC22    PC23   PC24    PC25    PC26    PC27    PC28
Standard deviation     0.16565 0.15602 0.1344 0.12442 0.09043 0.08307 0.03987
Proportion of Variance 0.00091 0.00081 0.0006 0.00052 0.00027 0.00023 0.00005
Cumulative Proportion  0.99749 0.99830 0.9989 0.99942 0.99969 0.99992 0.99997
                          PC29    PC30
Standard deviation     0.02736 0.01153
Proportion of Variance 0.00002 0.00000
Cumulative Proportion  1.00000 1.00000
```

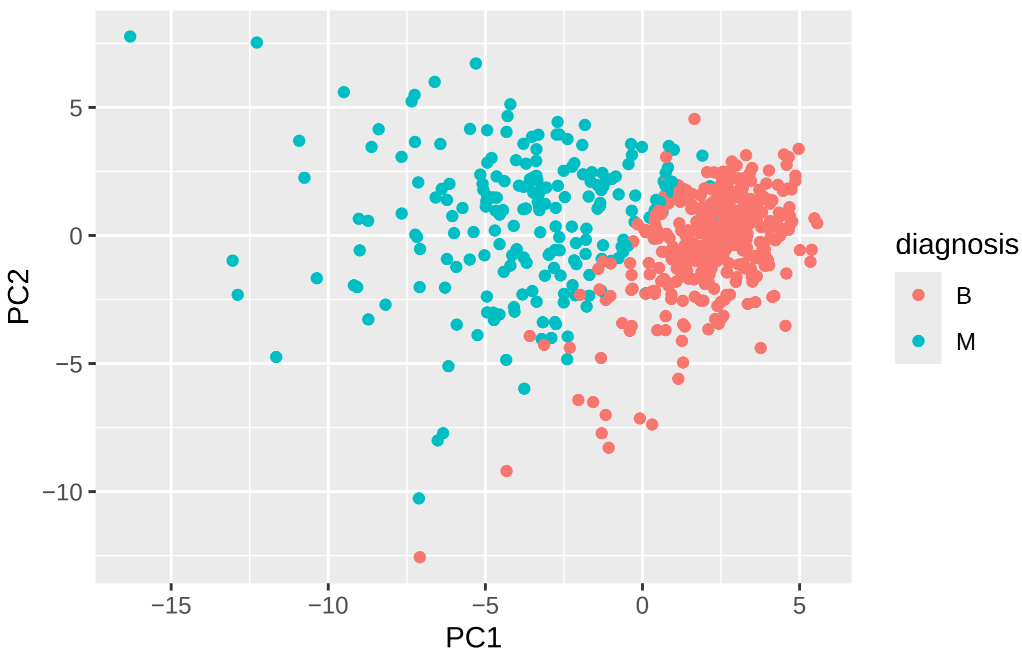
Q7. What stands out to you about this plot? Is it easy or difficult to understand? Why?

The plot is fairly easy to understand because the two diagnoses form noticeable clusters, especially along PC1. Benign and malignant samples are mostly separated, although there is some overlap. This suggests the first few principal components capture meaningful differences in the data.

Lets see the "PC score plot"

```
library(ggplot2)

ggplot(wisc.pr$x) + aes(PC1, PC2, col = diagnosis) + geom_point()
```



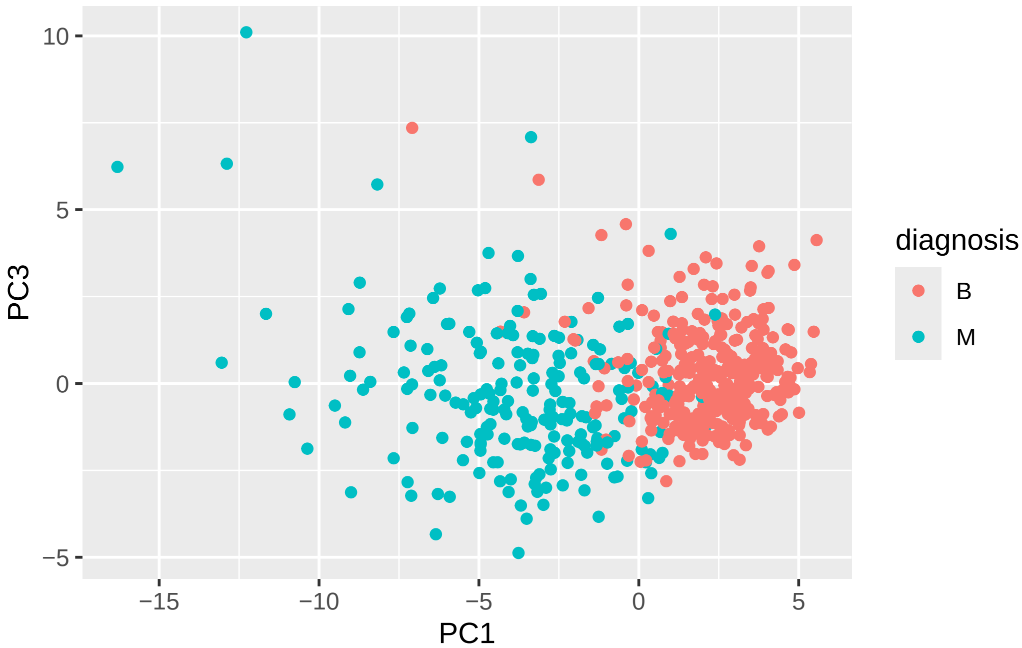
Q8. Generate a similar plot for principal components 1 and 3. What do you notice about these plots?

The PC1 vs PC3 plot still shows separation between benign and malignant samples, mainly along PC1. However, the groups overlap more compared to the PC1 vs PC2 plot, so PC3 is less useful for separating the diagnoses. This suggests PC2 captures more meaningful variation related to diagnosis than PC3.

Something I noticed about this plot is that it still shows separation between benign and malignant samples mainly along PC1. But the groups overlap more compared to the PC1 vs

PC2 pllot, so PC3 is less useful for separating the diagnoses. This suggest PC2 captures more meaningul variation related to diagnosis than PC3.

```
ggplot(wisc.pr$x) +
  aes(PC1, PC3, col = diagnosis) +
  geom_point()
```

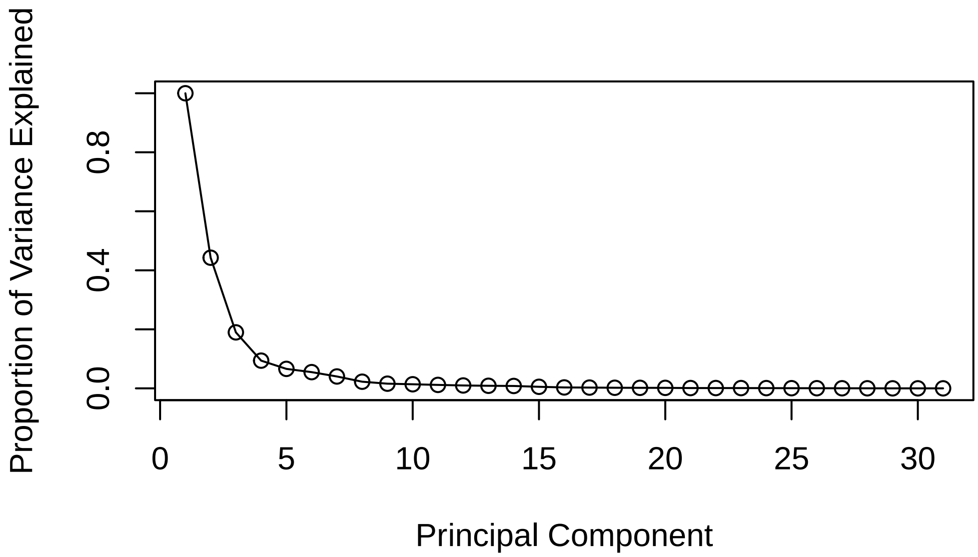



```
pr.var <- wisc.pr$sdev^2
head(pr.var)
```

```
[1] 13.281608  5.691355  2.817949  1.980640  1.648731  1.207357
```

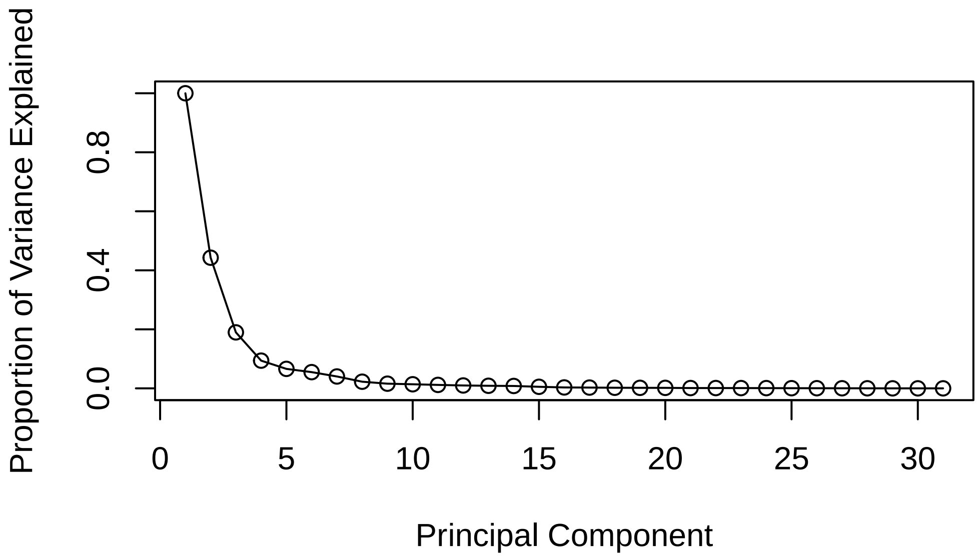
```
pve <- pr.var / sum(pr.var)

plot(c(1, pve),
     xlab = "Principal Component",
     ylab = "Proportion of Variance Explained",
     ylim = c(0,1), type = "o")
```




```r
pve <- pr.var / sum(pr.var)

plot(c(1,pve), xlab = "Principal Component",
     ylab = "Proportion of Variance Explained",
     ylim = c(0, 1), type = "o")
```

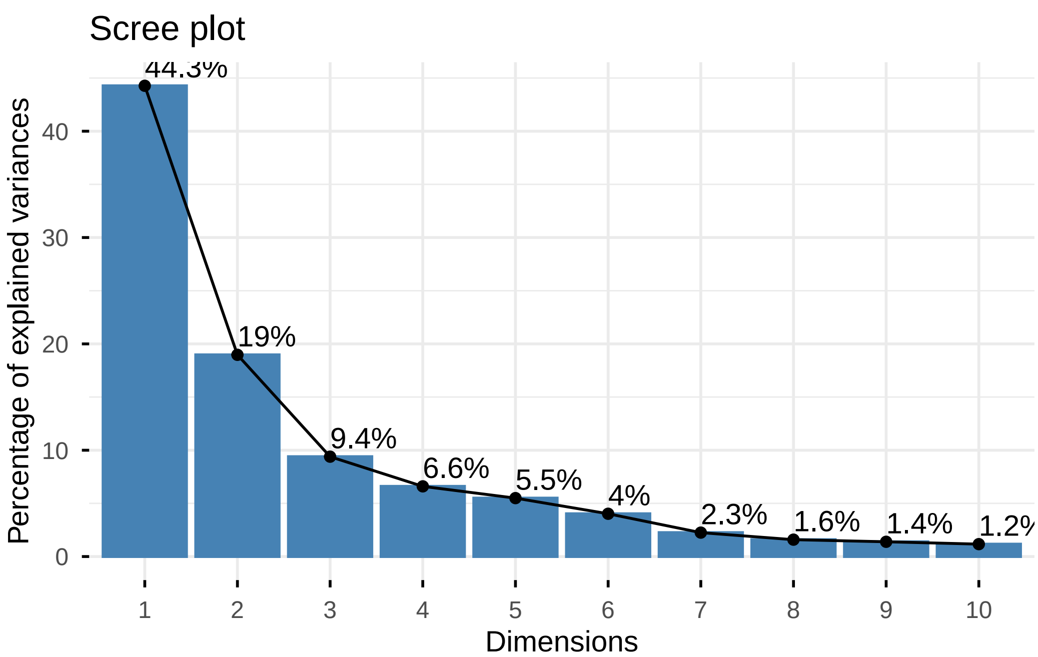



```r
#install.packages("factoextra")
library(factoextra)
```

```
Welcome to factoextra!

Want to learn more? See two factoextra-related books at https://www.datanovia.com/en/product/
guide-to-principal-component-methods-in-r/
```

```r
fviz_eig(wisc.pr, addlabels = TRUE)
```



> Q9. For the first principal component, what is the component of the loading vector (i.e. wisc.pr$rotation[,1]) for the feature concave.points_mean? This tells us how much this original feature contributes to the first PC. Are there any features with larger contributions than this one?

The loading value for concave.points_mean on the first principal component is approximately -0.261. Becaue of a negative value, this indicates direction, not importance. Its magnitude shows that it is a strong contributor to PC1, and only a few features have larger contributions.

```r
wisc.pr$rotation["concave.points_mean", 1]
```

```
[1] -0.2608538
```

```r
sort(abs(wisc.pr$rotation[,1]), decreasing = TRUE)[1:10]
```

```
 concave.points_mean       concavity_mean concave.points_worst 
           0.2608538            0.2584005            0.2508860 
    compactness_mean      perimeter_worst      concavity_worst 
           0.2392854            0.2366397            0.2287675 
        radius_worst       perimeter_mean           area_worst 
           0.2279966            0.2275373            0.2248705 
           area_mean 
           0.2209950 
```

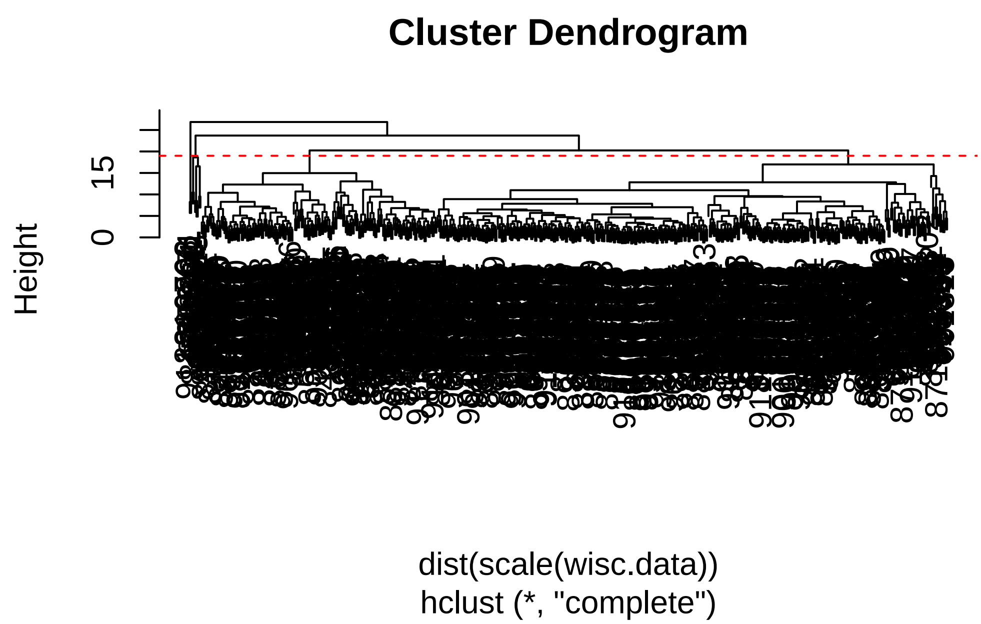
Hierchial Clustering

> Q10. Using the plot() and abline() functions, what is the height at which the clustering model has 4 clusters?

The clustering model has 4 clusters at approximately height 19.

```
wisc.hclust <- hclust( dist(scale(wisc.data)))
```

```
plot(wisc.hclust)
abline(h=19, col = "red", lty = 2)
```



Selecting Numbers of Clusters

```
wisc.hclust.clusters <- cutree(wisc.hclust, k = 4)
table(wisc.hclust.clusters, diagnosis)
```

```
                    diagnosis
wisc.hclust.clusters   B   M
                   1  12 165
```

```
2   2   5
3 343  40
4   0   2
```

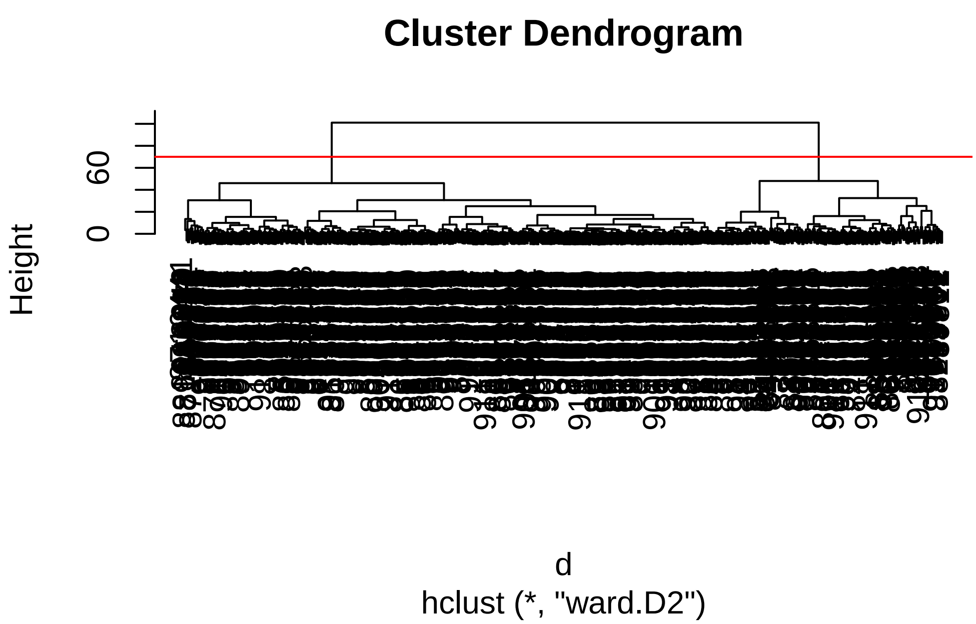
Q12. Which method gives your favorite results for the same data.dist dataset? Explain your reasoning.

I personally think usigng ward.D2 is my favorite. This is because it tends to create more balanced and clearly separated clusters. It minimizes variation within clusters, which often gives cleaner groupings than single or complete linkage. For this dataset, it appears to separate benign and malignant samples more effectively.

Combining Methods

class

```r
d <- dist(wisc.pr$x[,1:4])
wisc.pr.hclust <- hclust(d, method = "ward.D2")
plot(wisc.pr.hclust)
abline(h=70, col = "red")
```



```r
grps <- cutree(wisc.pr.hclust, h=70)
table(grps)
```

```
grps
  1   2 
171 398 
```

How does this clustering `grps` correspond to the expert `diagnosis`

```r
table(grps,diagnosis)
```

```
    diagnosis
grps   B   M
   1   6 165
   2 351  47
```

> Q13.How well does the newly created hclust model with two clusters separate out the two "M" and "B" diagnoses?

The newly created hclust model seperates the diagnoses very well. Cluster 1 contains mostly malignant samples (188M, 28B), while Cluster 2 contains mostly benign samples (329B, 24 M). This indicates the odel does a strong job to help distnguish between the two types of tumours, through a small number of samples are misclassfiied.

```r
wisc.pr.hclust.clusters <- cutree(wisc.pr.hclust, k=2)
table(wisc.pr.hclust.clusters, diagnosis)
```

```
                       diagnosis
wisc.pr.hclust.clusters   B   M
                      1   6 165
                      2 351  47
```

```r
table(wisc.pr.hclust.clusters, diagnosis)
```

```
                       diagnosis
wisc.pr.hclust.clusters   B   M
                      1   6 165
                      2 351  47
```

> Q14. How well do the hierarchical clustering models you created in the previous sections (i.e. without first doing PCA) do in terms of separating the diagnoses? Again, use the table() function to compare the output of each model (wisc.hclust.clusters and wisc.pr.hclust.clusters) with the vector containing the actual diagnoses.

The previous hierarchical clustering model without PCA separated the diagnoses moderatly well, but not as accurate as the PCA based model. Some clusters wer mostly malignant while another was mostly benign, but there was more mixing between benign and malignant samples. This suggests clustering after PCA gave clearer separation.

```r
table(wisc.hclust.clusters, diagnosis)
```

```
                    diagnosis
wisc.hclust.clusters   B   M
                   1  12 165
                   2   2   5
                   3 343  40
                   4   0   2
```

Sensitivity/ Specificity

> Q15. OPTIONAL: Which of your analysis procedures resulted in a clustering model with the best specificity? How about sensitivity?

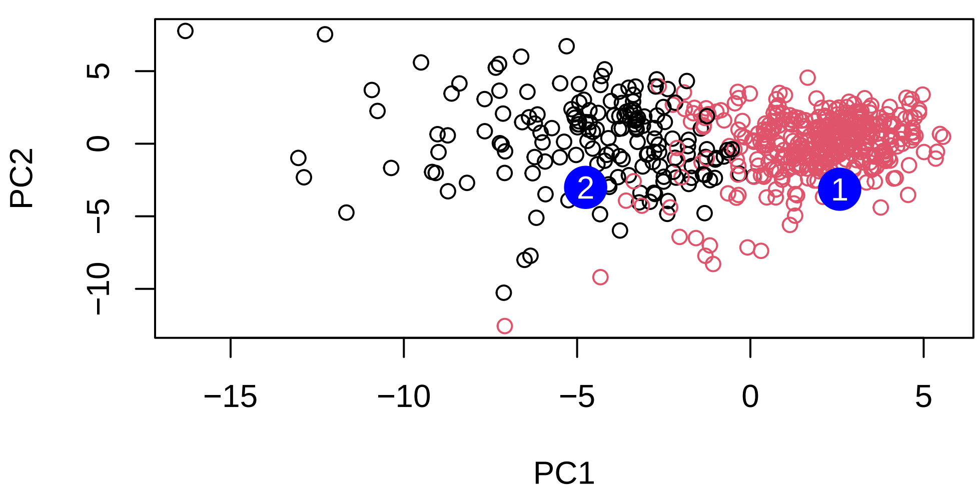
The clustering model using PCA before hierchial clustering resulted in the best specificity and sensitivity. It correctly identified most malignant samples, giving high sensitivity, and also correctly grouped most benign samples, giving high specificicity. The model witout PCA performed worse because more samples were misclassified.

```r
url <- "https://tinyurl.com/new-samples-CSV"
new <- read.csv(url)
npc <- predict(wisc.pr, newdata=new)
npc
```

```
           PC1       PC2        PC3        PC4       PC5        PC6        PC7
[1,]  2.576616 -3.135913  1.3990492 -0.7631950  2.781648 -0.8150185 -0.3959098
[2,] -4.754928 -3.009033 -0.1660946 -0.6052952 -1.140698 -1.2189945  0.8193031
            PC8       PC9       PC10      PC11      PC12      PC13     PC14
[1,] -0.2307350 0.1029569 -0.9272861 0.3411457  0.375921 0.1610764 1.187882
[2,] -0.3307423 0.5281896 -0.4855301 0.7173233 -1.185917 0.5893856 0.303029
          PC15       PC16        PC17        PC18        PC19       PC20
```

```
[1,] 0.3216974 -0.1743616 -0.07875393 -0.11207028 -0.08802955 -0.2495216
[2,] 0.1299153  0.1448061 -0.40509706  0.06565549  0.25591230 -0.4289500
           PC21        PC22        PC23        PC24         PC25         PC26
[1,]  0.1228233 0.09358453 0.08347651  0.1223396  0.02124121  0.078884581
[2,] -0.1224776 0.01732146 0.06316631 -0.2338618 -0.20755948 -0.009833238
             PC27         PC28         PC29         PC30
[1,]  0.220199544 -0.02946023 -0.015620933  0.005269029
[2,] -0.001134152  0.09638361  0.002795349 -0.019015820
```

```r
plot(wisc.pr$x[,1:2], col=grps)
points(npc[,1], npc[,2], col="blue", pch=16, cex=3)
text(npc[,1], npc[,2], c(1,2), col="white")
```



Q16. Which of these new patients should we prioritize for follow up based on your results?

Patient 2. should be prioritized for follow up because it falls closer to the cluster associated with malignant tumors, while patient 1 is located more with benign cluster. Because Patient 2 has a higher risk of malignancy they should recieve testing first.